RACER-GT User Guide

Installation, Workflow, Data Contracts, and Outputs

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2026-07-27 • version 1.1.0

Abstract

This guide takes a researcher from installation to an accepted daily Google Trends index. It documents the data contract a collector must satisfy, the four command-line steps, the Python API, every output file, and the failure modes that are most likely to occur in practice. Method and proofs are in the methodology manuscript and the mathematical appendix; this document is about operating the software correctly.

Contents

1 Installation

RACER-GT requires Python 3.10 or later.

```
# from PyPI
python -m pip install racergt

# from a release wheel
python -m pip install racergt-1.0.0-py3-none-any.whl

# from source, for development
python -m pip install -e '.[dev]'
pytest -q
```

The package is deliberately *ingestion-first*: it contains no Google Trends scraper. Endpoints, rate limits, authentication, and terms of service change, and a statistical method that depends on an unofficial client inherits that fragility. The collector is your choice; the package only requires that it emit the schema in section 3.

2 The four-step workflow

2.1 Step 1: lock the protocol

```
racergt design config.yaml --out protocol_bundle --anchor-date 2026-08-01
```

This produces `protocol.lock.yaml` with a SHA-256 protocol hash, `collection_schedul` with the balanced day × stream × replicate plan, `chunk_windows.csv` with the fixed query windows, and an audit manifest.

Remark 2.1 (Do not hand-edit the lock file). `RacerGTConfig.load_yaml` *recomputes the hash and raises if it disagrees with the stored value. Editing a locked setting by hand therefore fails loudly, which is the intended behaviour. To change a setting, edit the source configuration and regenerate.*

2.2 Step 2: collect and convert

Follow the schedule. Preserve every raw response unmodified, then convert to the long format of section 3. Record a hash of both the raw response and the numeric vector for each chunk. If a network interruption, login change, cookie clear, or client update occurs, record it in the metadata; do not decide unilaterally that the pull is invalid, because that decision would be a value-dependent exclusion.

2.3 Step 3: audit

```
racergt audit config.yaml raw_chunks.csv --out audit.json
```

The audit checks the batch against the locked protocol before any estimation happens. It exits with status 2 if the batch fails.

2.4 Step 4: run

```
racergt run config.yaml raw_chunks.csv \
  --benchmark lower_frequency_benchmark.csv \
  --out results
```

Exit status is 3 if the locked decision tree returns FAIL, so the command can gate an automated pipeline.

2.5 Python API

```
import pandas as pd
from racergt import RacerGTConfig, RacerGTPipeline

config = RacerGTConfig.load_yaml("config.yaml")
raw = pd.read_csv("raw_chunks.csv")
benchmark = pd.read_csv("lower_frequency_benchmark.csv")

result = RacerGTPipeline(config).fit(raw, benchmark=benchmark)
result.save("results")

print(result.decision.status)      # PASS / REVIEW / FAIL
print(result.final_series.head())
```

3 The data contract

3.1 Raw chunk table

One row per `pull_id` × `chunk_id` × `historical_date`.

Table 1: Required columns.

Column	Type	Meaning
`series_id`	string	research indicator identifier
`pull_id`	string	one complete history reconstruction
`chunk_id`	string	fixed query window, must exist in the manifest
`historical_date`	date	GT observation date
`value`	float	raw GT relative search value, normally 0–100
`collection_day`	integer	design level of the collection day
`stream_id`	string	fixed composite collection environment
`replicate_id`	string	technical replicate within a day × stream cell
`window_start`	date	chunk start
`window_end`	date	chunk end

Remark 3.1 (Two columns that are routinely misunderstood). `collection_day` *is the* ordinal design level *of the collection day (0, 1, 2, ...), not the day offset and not a calendar date. And* `stream_id` *denotes a composite environment—device, browser profile, cookies, account state, network route, IP. An estimated stream effect must not be reported as an IP effect unless a crossover factorial design was actually run.*

Strongly recommended additional columns: `retrieved_at`, `protocol_hash`, `keyword`, `geo`, `category`, `search_property`, `language`, `is_partial`, `raw_response_hash`, `network_event`.

3.2 Lower-frequency benchmark table

Requires `series_id`, `period_start`, `period_end`, `value`; `se` is optional and falls back to the protocol default.

4 Outputs

Table 2: Files written by `result.save()`.

Path	Contents
`RACER_GT_final_series.csv`	final daily index, conditional SE, 95% interval
`complete_calibrated_pulls.csv`	every calibrated pull
`calibration/*_chunk_scales.csv`	per-chunk global scale estimates
`calibration/*_overlap_edges.csv`	robust edge estimates and diagnostics
`duplicates/`	exact groups, pair metrics, components, residuals
`consensus/design_weighted_consensus.csv`	design-based reference estimator
`consensus/consensus_weights.csv`	GLS / minimum-variance weights
`gstudy/`	ANOVA, variance components, D-study, per transformation
`reliability/`	pairwise reliability, convergence, day/stream dependence
`decision/acceptance_decision_tree.csv`	every rule, observed vs threshold
`report/RACER_GT_diagnostic_report.md`	figure-based diagnostic report
`summary.json`	protocol hash, status, headline diagnostics

Remark 4.1 (Two consensus files, one final series). `consensus/design_weighted_consensus` *is a design-based reference estimator. It is reported for comparison and does* not *feed* `RACER_GT_final_series.csv`, *which comes from the temporal benchmark when one is supplied and from the GLS consensus otherwise. Large disagreement between the two is a warning sign worth investigating, not a bug.*

5 Reading the decision

A batch is FAIL if any mandatory rule fails, REVIEW if none fails but at least one is indeterminate, and PASS otherwise. Common causes, and what to do:

Table 3: Frequent outcomes and their remedies.

Symptom	Interpretation and action
`OverlapGraphError`	the chunk design leaves a gap. Increase window length or reduce step so that consecutive chunks share at least `min_overlap_days` usable days.
Low spectral effective pulls	retrievals are highly dependent. Add collection days or streams; adding technical replicates will not help.
Detection rule indeterminate	no zero/non-zero variation, so κ is undefined. The rule is automatically non-mandatory in this case and needs no action.
High zero share	the term is too rare for daily frequency. Move to weekly, or use a broader Topic rather than a Term.
Innovation G below threshold, level G above	levels reproduce but daily changes do not. Do not use the series in differences.
Benchmark RMSE too large	the daily consensus and the lower-frequency series disagree. Check that both use the identical query specification.

6 Stata interoperability

Every table can be exported to Stata 118 format:

```
racergt export-stata results/RACER_GT_final_series.csv final_series.dta
```

Supported read formats are CSV, Parquet, and XLS/XLSX; supported write formats are CSV, Parquet, XLSX, and Stata 118 `.dta`.

7 Research conduct requirements

These are not optional style preferences; violating them invalidates the guarantees the framework provides.

- A static IP address is not evidence of an independent sample.
- Do not adjust reliability thresholds or select pulls after seeing financial results.
- If pilot monitoring leads to any protocol or decision-rule change, the monitored data become pilot data and the confirmatory batch restarts.

- Retain exact duplicates in the audit archive.
- PASS means the batch satisfies the locked measurement rules. It does not establish construct validity for the keyword and licenses no causal claim.